

Integral lattices

Henry Twiss

1 Integral Lattices

Let F be a characteristic zero field and V be an F -vector space of dimension n with nondegenerate symmetric inner product $\langle \cdot, \cdot \rangle$. We say that a subset Λ of V is a **integral lattice** if Λ is a free $\mathbb{Z}$ -module. In particular, any integral lattice Λ is of the form

$$\Lambda = \mathbb{Z}v_1 + \cdots + \mathbb{Z}v_m,$$

for some linearly independent vectors $v_1, \dots, v_m$ of V with $m \leq n$. We say Λ is **complete** if $n = m$. This means $v_1, \dots, v_n$ is necessarily a basis of V . If $e_1, \dots, e_n$ is an orthonormal basis for V , write

$$v_i = \sum_j v_{i,j} e_j,$$

with $v_{i,j} \in F$, and define the associated **generator matrix** P by

$$P = \begin{pmatrix} v_{1,1} & \cdots & v_{n,1} \\ \vdots & & \vdots \\ v_{1,n} & \cdots & v_{n,n} \end{pmatrix}.$$

Then P is the base change matrix from $e_1, \dots, e_n$ to $v_1, \dots, v_n$. The **covolume** V_Λ of a complete integral lattice Λ is defined to be

$$V_\Lambda = |\det(P)|.$$

As the base change matrix between any two orthonormal bases is an orthogonal matrix and hence has determinant ± 1 , the covolume is independent of the choice of bases. In the case where V is a real or complex vector space, we will always take $e_1, \dots, e_n$ to be the standard basis.

If Λ is a complete integral lattice in V then the **dual** $\Lambda^\vee$ of Λ is defined by

$$\Lambda^\vee = \{v \in V : \langle \lambda, v \rangle \in \mathbb{Z} \text{ for all } \lambda \in \Lambda\}.$$

In other words, the dual integral lattice consists of all of the vectors in V whose inner product with elements of the integral lattice are integers. Clearly the dual integral lattice is an abelian group. Less clear is that the dual integral lattice turns out to always be complete.

Proposition 1.1. *If $v_1, \dots, v_n$ is a basis for a complete integral lattice Λ then the dual basis $v_1^\vee, \dots, v_n^\vee$ is a basis for $\Lambda^\vee$. In particular, $\Lambda^\vee$ is a complete integral lattice.*

Proof. Let $v \in V$ and write

$$v = \sum_j a_j v_j^\vee,$$

with $v_j^\vee \in \mathbb{R}$. Since the dual basis is determined by $\langle v_i, v_j^\vee \rangle = \delta_{i,j}$, it follows that $v \in \Lambda^\vee$ if and only if $v_j^\vee \in \mathbb{Z}$. This means $v_1^\vee, \dots, v_n^\vee$ is a basis for $\Lambda^\vee$ as a free $\mathbb{Z}$ -module and thus is a complete integral lattice. $\square$

Since the dual of the dual basis is the original basis, $(\Lambda^\vee)^\vee = \Lambda$. We say that Λ is **self-dual** if $\Lambda^\vee = \Lambda$. For example, the integral lattice $\mathbb{Z}^n$ is self-dual because the standard basis is self-dual. It also turns out that the covolume of the dual integral lattice is the inverse of the volume of the original integral lattice.

Proposition 1.2. *Let Λ be a complete integral lattice in V . Then*

$$V_{\Lambda^\vee} = \frac{1}{V_\Lambda}.$$

Proof. Let $e_1, \dots, e_n$ be an orthonormal basis for V , let $v_1, \dots, v_n$ be a basis for Λ , and let P be the associated generator matrix. By Proposition 1.1, $v_1^\vee, \dots, v_n^\vee$ is a basis for $\Lambda^\vee$ and $(P^{-1})^t$ is the base change matrix from $e_1, \dots, e_n$ to $v_1^\vee, \dots, v_n^\vee$. As $\det((P^{-1})^t) = \det(P)^{-1}$ the claim follows. $\square$

We now turn to the case when V is a real inner product space of dimension n . Let $d\lambda$ be the Lebesgue measure induced by the inner product with associated volume Vol given by

$$\text{Vol}(M) = \int_M d\lambda,$$

for any measurable set M . Note that Λ acts on V by automorphisms given by translation. In other words, we have a group action

$$\Lambda \times V \rightarrow V \quad (\lambda, v) \mapsto \lambda + v.$$

Moreover, Λ acts properly discontinuously on V . To see this, let $v \in V$ and let δ_v be such that $0 < \delta_v < \min_i(v - v_i)$. Taking U_v to be the ball of radius δ_v about v , the intersection $\lambda + U_v \cap U_v$ is empty unless $\lambda = 0$. As Λ is also discrete, it follows that V/Λ is also connected Hausdorff. Whence V/Λ admits a fundamental domain

$$\mathcal{M} = \{t_1 v_1 + \dots + t_n v_n \in V : 0 \leq t_i \leq 1 \text{ for all } i\}.$$

Moreover, any translation of $\mathcal{M}$ by an element of Λ is also a fundamental domain. As we might expect, the covolume of Λ is equal to the volume of $\mathcal{M}$.

Proposition 1.3. *Let Λ be a complete integral lattice in a finite dimensional real inner product space V and let $\mathcal{M}$ be a fundamental domain for Λ . Then*

$$V_\Lambda = \text{Vol}(\mathcal{M}).$$

Proof. Let $e_1, \dots, e_n$ be an orthonormal basis of V , and $v_1, \dots, v_n$ be a basis for Λ . Also let P be the associated generator matrix. The change of variables

$$x_1 e_1 + \dots + x_n e_n \mapsto x_1 v_1 + \dots + x_n v_n,$$

has Jacobian determinant V_Λ . Whence

$$\text{Vol}(\mathcal{M}) = \int_{\mathcal{M}} d\lambda = V_\Lambda \int_{[0,1]^n} d\mathbf{x} = V_\Lambda. \quad \square$$

This result shows that the covolume of a complete integral lattice in a real inner product space is a measure of the density of the integral lattice. The smaller the covolume the smaller the fundamental domain and the more dense the integral lattice is.

It turns out that for a real inner product space, being an integral lattice is equivalent to being a discrete subgroup.

Proposition 1.4. *Let Λ be a subset of a finite dimensional real inner product space V . Then Λ is an integral lattice if and only if it is a discrete subgroup.*

Proof. It is clear that if Λ is an integral lattice then it is a discrete subgroup. So suppose Λ is a discrete subgroup. Then Λ is closed. Let V' be the subspace spanned by Λ and let its dimension be m . Choosing a basis $v'_1, \dots, v'_m$ of V' , set

$$\Lambda' = \mathbb{Z}v'_1 + \dots + \mathbb{Z}v'_m.$$

Then Λ' is a complete integral lattice in V' and $\Lambda' \subseteq \Lambda \subset V$. We claim Λ/Λ' is finite. To see this, let λ be a representative of a coset in Λ/Λ' and let $\mathcal{M}'$ be a fundamental domain for Λ' in V' . As $\mathcal{M}'$ is a fundamental domain, there exists unique $w' \in \mathcal{M}'$ and $\lambda' \in \Lambda'$ such that $\lambda = w' + \lambda'$. But then $w' = \lambda - \lambda' \in \mathcal{M}' \cap \Lambda$ and since $\mathcal{M}' \cap \Lambda$ is closed, discrete, and compact, it must be finite. Hence there are finitely many w' and thus finitely many cosets in Λ/Λ' . Letting $|\Lambda/\Lambda'| = q$, we have $q\Lambda \subseteq \Lambda'$ and therefore

$$\Lambda \subseteq \Lambda' = \mathbb{Z}\frac{1}{q}v'_1 + \dots + \mathbb{Z}\frac{1}{q}v'_m.$$

In particular, Λ is a subgroup of a free $\mathbb{Z}$ -module and therefore is a free $\mathbb{Z}$ -module itself. This means Λ is an integral lattice. $\square$

As for complete integral lattices in V , they are equivalent to the existence of a bounded set whose translates cover V .

Proposition 1.5. *Let Λ be an integral lattice in a finite dimensional real inner product space V . Then Λ is complete if and only if there exists a bounded subset M of V whose translates by Λ cover V .*

Proof. First suppose Λ is complete. Then we may take $M = \mathcal{M}$ to be a fundamental domain of Λ which is bounded and whose translates by Λ cover V . Now suppose Λ is an integral lattice and there exists a bounded subset M of V whose translates by Λ cover V . Let W be

the subspace of V spanned by Λ . Then W is closed. Moreover, Λ is complete if and only if $V = W$ and this is what we will show. So let $v \in V$. Since the translates of M by Λ cover V , for every positive integer n we may write

$$nv = w_n + \lambda_n,$$

with $v_n \in M$ and $\lambda_n \in \Lambda$. As M is bounded, $\lim_{n \rightarrow \infty} \frac{1}{n}v_n = 0$. Moreover, $\frac{1}{n}\lambda_n \in W$ and since W is closed we must have that $\lim_{n \rightarrow \infty} \frac{1}{n}\lambda_n$ exists and belongs to W . These two limits together imply

$$v = \lim_{n \rightarrow \infty} \frac{1}{n}\lambda_n,$$

and is an element of W . Thus $V = W$ which means Λ is complete. $\square$

The most important result we will require about integral lattices is **Minkowski's lattice point theorem** which states that, under some mild conditions, a set of sufficiently large volume in V contains a nonzero point of a complete integral lattice.

Theorem (Minkowski's lattice point theorem). *Suppose Λ is a complete integral lattice in a real inner product space V of dimension n . Let $X \subset V$ is a compact convex symmetric set. If*

$$\text{Vol}(X) \geq 2^n V_\Lambda,$$

then there exists a nonzero $\lambda \in \Lambda \cap X$.

Proof. We will prove the claim depending on if the inequality is strict or not. First suppose $\text{Vol}(X) > 2^n V_\Lambda$. Consider the linear map

$$\phi : \frac{1}{2}X \rightarrow V/\Lambda \quad \frac{1}{2}x \mapsto \frac{1}{2}x + \Lambda.$$

If ϕ were injective then we would have

$$\text{Vol}\left(\frac{1}{2}X\right) = \frac{1}{2^n} \text{Vol}(X) \leq V_\Lambda,$$

so that $\text{Vol}(X) \leq 2^n V_\Lambda$. This is a contradiction, so ϕ cannot be injective. Hence there exists distinct $x_1, x_2 \in \frac{1}{2}X$ such that $\phi(x_1) = \phi(x_2)$. Thus $2x_1, 2x_2 \in X$. In particular, since X is symmetric we must have $-2x_2 \in X$. Convexity of X implies

$$\left(1 - \frac{1}{2}\right)2x_1 + \frac{1}{2}(-2x_2) = x_1 - x_2,$$

is an element of X . But $x_1 - x_2 \in \Lambda$ because $\phi(x_1) = \phi(x_2)$ and ϕ is linear. Then $\lambda = x_1 - x_2$ is nonzero with $\lambda \in \Lambda \cap X$. Now suppose $\text{Vol}(X) = 2^n V_\Lambda$. Then

$$\text{Vol}((1 + \varepsilon)X) = (1 + \varepsilon)^n 2^n V_\Lambda > 2^n V_\Lambda.$$

What we have just proved shows that there exists a nonzero $\lambda_\varepsilon \in \Lambda \cap (1 + \varepsilon)X$. In particular, if $\varepsilon \leq 1$ then $\lambda_\varepsilon \in 2X \cap \Lambda$. The set $2X \cap \Lambda$ is compact and discrete and therefore finite.

Taking $\varepsilon = \frac{1}{n}$, the sequence $(\lambda_{\frac{1}{n}})_{n \geq 1}$ belongs to the finite set $2X \cap \Lambda$ and so must converge to a point λ . Since Λ is discrete and the $\lambda_{\frac{1}{n}}$ are nonzero so too is λ . As λ is an element of

$$\bigcap_{n \geq 1} \left(1 + \frac{1}{n}\right) X,$$

and X is closed, $\lambda \in X$ as well. Thus we have found a nonzero $\lambda \in \Lambda \cap X$. □